

## Research Summary

The long-term goal of my research is to illuminate the biophysical mechanisms of gene regulation through the high-throughput measurement and quantitative modeling of sequence-function relationships. My lab pursues this goal using a combination of experimental, computational, and mathematical approaches. Our experimental work develops high-throughput mutagenesis assays for measuring sequence-function relationships in alternative mRNA splicing (in humans), transcriptional regulation (in bacteria), and other biological systems. Our computational work develops methods for quantitatively modeling data from high-throughput mutagenesis assays, as well as for analyzing the sequence-function relationships described by genomic AI models. Our mathematical work addresses fundamental questions about the quantitative nature of sequence-function relationships and their statistical inference from high-throughput data.

## General information

Name: Justin Block Kinney, PhD  
Address: 2116 Koch Building, Cold Spring Harbor Laboratory,  
1 Bungtown Rd., Cold Spring Harbor, NY 11724  
Phone: 631-897-5731  
Email: [jkinney@cshl.edu](mailto:jkinney@cshl.edu)  
Website: <https://jbkinney.github.io/>  
Nationality: United States Citizen

## Employment and education

2025-present **Professor, Simons Center for Quantitative Biology, Cold Spring Harbor Laboratory**

2020-2025 **Associate Professor, Simons Center for Quantitative Biology, Cold Spring Harbor Laboratory**

2014-2020 **Assistant Professor, Simons Center for Quantitative Biology, Cold Spring Harbor Laboratory**

2010-2014 **Fellow, Simons Center for Quantitative Biology, Cold Spring Harbor Laboratory**

2009-2010 **Postdoctoral Fellow, Simons Center for Quantitative Biology, Cold Spring Harbor Laboratory**  
Mentors: Bruce Stillman and Michael Zhang

2008-2009 **Postdoctoral Fellow, Department of Physics, Princeton University**  
Mentors: Curtis Callan and Edward Cox

2002-2008 **Graduate Student, Department of Physics, Princeton University**  
**PhD in Physics, September 2008**  
Experimental Biophysics under Edward Cox, 2007-2008  
Theoretical Biophysics under Curtis Callan, 2004-2008  
Theoretical High Energy Physics under Juan Maldacena, 2003-2004

1998-2002 **Undergraduate Student, Departments of Physics and Math, Cornell University**  
BA in Physics, summa cum laude, May 2002  
BA in Mathematics, magna cum laude, May 2002

## Professional activities at CSHL

2025-present Chair, Simons Center for Quantitative Biology, CSHL  
 2023-present Co-leader, Gene Regulation and Inheritance Program, CSHL Cancer Center  
 2020-2024 Special Advisor in Biostatistics, CSHL Cancer Center  
 2020-present Lead Instructor, Biostatistics Course, CSHL Cancer Center  
 2017-present Lead instructor in Quantitative Biology, CSHL School of Biological Sciences  
 2016-present Lead Instructor, URP Python Programming Course, CSHL School of Biological Sciences  
 2013-present Affiliate Member, [bioRxiv](#)  
 2010-present Instructor, CSHL School of Biological Sciences

## Awards and Honors

2015 [Inaugural Info-Metrics Prize](#), Info-Metrics Institute at American University  
 2014 [Scialog Fellow](#), Moore Foundation / Research Corporation for Science Advancement  
 2005 [Outstanding Teaching Assistant](#), American Association of Physics Teachers  
 2005 Teaching Prize, Princeton University Physics Department  
 2002 Van Zandt Williams, Sr. \*41 Fellowship, Princeton University  
 2002 Yennie Prize in Physics, Cornell University  
 2002 [Merrill Presidential Scholar](#), Cornell University  
 2002 Phi Beta Kappa  
 2001 [Barry M. Goldwater Scholarship](#)

## Patent applications

- Vakoc CH, Shi J, Kinney JB. Methods of identifying essential protein domains. US Patent App. 15/546, 206, #15546106 (2018).

## Preprints

1. Liu Z, Cordero A, Kinney JB. **PoolParty: streamlined design of DNA sequence libraries in Python.** [bioRxiv doi:10.64898/2026.04.06.716802](#) (2026).
2. Schilder BM, Liu Z, Desmarais JJ, Laub D, Rahimi F, Sethi P, Pereira LA, Sun M, Kinney JB, McCandlish DM, Zhou J, Koo PK. Genetic background shapes AI-predicted variant effects. [bioRxiv doi:10.64898/2026.04.04.715328](#) (2026).
3. Bauer J, Zali N, Chouhan OP, El Demerdash O, Loell K, Kinney JB, Joshua-Tor LW, Stillman B. Evolution of origin sequence and recognition for licensing of eukaryotic DNA replication. [bioRxiv doi:10.64898/2026.03.10.710760](#) (2026).
4. Martí-Gómez C, McCandlish DM, Kinney JB. **GaugeFixer: overcoming parameter non-identifiability in models of sequence-function relationships.** [bioRxiv doi:10.64898/2025.12.08.693054](#) (2025).
5. Seitz EE, McCandlish DM, Kinney JB\*, Koo PK\*. **Uncovering the Mechanistic Landscape of Regulatory DNA with Deep Learning.** [bioRxiv doi:10.1101/2025.10.07.681052](#) (2025). \*Corresponding authors.
6. Steinberg JI, Sertznig H, Desmarais JJ, Wilken J, Rubio D, Peacey M, Kinney JB, Schorn AJ. HENMT1 restricts endogenous retrovirus activity by methylation of 3'-tRNA fragments. [bioRxiv doi:10.1101/2025.05.12.650695](#) (2025).

## Research Publications

1. **Kinney JB.** Mount Fuji's stubby peak: the genotypic density of additive landscapes near maximal fitness. [Genetics](#) **iyag133** (2026).
2. Wu P, Wang J, Pryor B, Valentino I, Ritter DF, Loell K, Yarychkivska O, Shaham S, Kinney JB, Ercan S, Joshua-Tor L, Hammell CM. A molecular timer couples organism-wide temporal identity to developmental checkpoints. [Proc Natl Acad Sci USA](#) **123(19):e2606846123** (2026).
3. Petti S, Martí-Gómez C, Kinney JB, Zhou J, McCandlish DM. On learning functions over biological sequence space: relating Gaussian process priors, regularization, and gauge fixing. [J Math Biol](#) **92:78** (2026).
4. Martí-Gómez C, Zhou J, Chen W-C, Kinney JB, McCandlish DM. Inference and visualization of complex genotype-phenotype maps. [Mol Biol Evol](#) **msag023** (2026).
5. Alpsoy A, Ipsaro JJ, Skopelitis D, Pal S, Chung FS, Carpenter S, Desmarais JJ, Wu XS, Chang K, DiMare MT, Harten E, Bergman S, Kinney JB, Engelman JA, Bhang HC, Joshua-Tor L, Vakoc C. Structural Basis of DNA-Dependent Coactivator Recruitment by the Tuft Cell Master Regulator POU2F3. [Cell Rep](#) **44(11):116572** (2025).
6. Rousseau R, Kinney JB. Algebraic and diagrammatic methods for the rule-based modeling of multi-particle complexes. [PRX Life](#) **3, 023004** (2025).
7. Posfai A, McCandlish DM, Kinney JB. Symmetry, gauge freedoms, and the interpretability of sequence-function relationships. [Phys Rev Res](#) **7:023005** (2025).
8. Posfai A, Zhou J, McCandlish DM\*, Kinney JB\*. Gauge fixing for sequence-function relationships. [PLoS Comput Biol](#) **21(3):e1012818** (2025). \*Corresponding authors.
9. Seitz EE, McCandlish DM, Kinney JB\*, Koo PK\*. Interpreting cis-regulatory mechanisms from genomic deep neural networks using surrogate models. [Nat Mach Intell](#) doi:10.1038/s42256-024-00851-5 (2024). \*Corresponding authors.
10. Ishigami Y, Wong MS, Martí-Gómez C, Ayaz A, Kooshkbaghi M, Hanson S, McCandlish DM, Krainer AR\*, Kinney JB\*. Specificity, synergy, and mechanisms of splice-modifying drugs. [Nat Commun](#) **15(1):1880** (2024). \*Corresponding authors.
11. Pukhrambam C, Molodtsov V, Kooshkbaghi M, Tareen A, Vu H, Skalenko KS, Su M, Zhou Y, Winkelman JT, Kinney JB, Ebright, RH, Nickels BE. Structural and mechanistic basis of  $\sigma$ -dependent transcriptional pausing. [Proc Natl Acad Sci USA](#) **119(23):e2201301119** (2022).
12. Tareen A, Posfai A, Ireland WT, McCandlish DM, Kinney JB. MAVE-NN: learning genotype-phenotype maps from multiplex assays of variant effect. [Genome Biol](#) **23:98** (2022).
13. Zhou J, Wong MS, Chen W-C, Krainer AR, Kinney JB, McCandlish DM. Higher-order epistasis and phenotypic prediction. [Proc Natl Acad Sci USA](#) **119(39):e2204233119** (2022).
14. Chen W-C, Zhou J, Sheltzer JM, Kinney JB, McCandlish DM. Field-theoretic density estimation for biological sequence space with applications to 5' splice site diversity and aneuploidy in cancer. [Proc Natl Acad Sci USA](#) **118(40):e2025782118** (2021).
15. Skalenko KS, Li L, Zhang Y, Vvedenskaya IO, Winkelman JT, Cope A, Taylor DM, Shah P, Ebright RH, Kinney JB, Zhang Y, Nickels BE. Promoter-sequence determinants and structural basis of primer-dependent transcription initiation in *Escherichia coli*. [Proc Natl Acad Sci USA](#) **118(27):e2106388118** (2021).
16. Tareen A, Kinney JB. Logomaker: beautiful sequence logos in Python. [Bioinformatics](#) **36:2272-2274** (2020).

17. Ireland WT, Beeler SM, Flores-Bautista E, McCarty NS, Röschinger T, Belliveau NM, Sweredoski MJ, Moradian A, Kinney JB, Phillips R. Deciphering the regulatory genome of *Escherichia coli*, one hundred promoters at a time. [eLife 9:e55308](#) (2020).
18. Hu Y, Tareen A, Sheu Y-J, Ireland WT, Speck C, Li H, Joshua-Tor L, Kinney JB, Stillman B. Evolution of DNA replication origin specification and gene silencing mechanisms. [Nat Commun 11:5175](#) (2020).
19. **Tareen A, Kinney JB. Biophysical models of cis-regulation as interpretable neural networks. Proceedings of the 14th conference on Machine Learning in Computational Biology. [bioRxiv doi:10.1101/835942](#) (2019)**
20. Barnes SL, Belliveau NM, Ireland WT, Kinney JB, Phillips R. Mapping DNA sequence to transcription factor binding energy in vivo. [PLoS Comp Biol 15\(2\):e1006226](#) (2019).
21. Adams RM, Kinney JB, Walczak AM, Mora T. Epistasis in a fitness landscape defined by antibody-antigen binding free energy. [Cell Syst 8\(1\):86-93](#) (2019).
22. **Forcier T, Ayaz A, Gill MS, Jones D, Phillips R, Kinney JB. Measuring cis-regulatory energetics in living cells using allelic manifolds. [eLife 7:e40618](#) (2018).**
23. Lu B, Klingbeil O, Tarumoto Y, Somerville TDD, Huang Y-H, Wei Y, Wai DC, Low JKK, Milazzo JP, Wu XS, Cao Z, Yan X, Demerdash OE, Huang G, Mackay JP, Kinney JB, Shi J, Vakoc CR. A Transcription Factor Addiction in Leukemia Imposed by the MLL Promoter Sequence. [Cancer Cell 34\(6\):970-978](#) (2018).
24. **Chen W, Tareen A, Kinney JB. Density estimation on small datasets. [Phys Rev Lett 121\(16\):160605](#) (2018).**
25. **Wong MS, Kinney JB\*, Krainer AR\*. Quantitative activity profile and context dependence of all human 5' splice sites. [Mol Cell 71\(6\):1012-1026.e3](#) (2018).**  
*\*Corresponding authors.*
26. Posfai A, Zhou J, Plotkin JB, Kinney JB, McCandlish DM. Selection for protein stability enriches for epistatic interactions. [Genes 9\(9\):423](#) (2018).
27. Belliveau NM, Barnes SL, Ireland WT, Jones DJ, Sweredoski MJ, Moradian A, Hess S, Kinney JB, Phillips R. A systematic approach for dissecting the molecular mechanisms of transcriptional regulation in bacteria. [Proc Natl Acad Sci USA 115\(21\):E4796-E4805](#) (2018).
28. Ipsaro JJ, Shen C, Eri A, Xu Y, Kinney JB, Joshua-Tor L, Vakoc CR, Shi J. Rapid generation of drug-resistance alleles at endogenous loci using CRISPR-Cas9 indel mutagenesis. [PLoS ONE 12\(2\):e0172177](#) (2017).
29. Senturk S, Shirole NH, Nowak DD, Corbo V, Pal D, Vaughan A, Tuveson DA, Trotman LC, Kepecs A, Kinney JB, Sordella R. Rapid and tunable method to temporally control gene editing based on conditional Cas9 stabilization. [Nat Commun 8:14370](#) (2017).
30. **Adams RM, Mora T\*, Walczak AM\*, Kinney JB\*. Measuring the sequence-affinity landscape of antibodies with massively parallel titration curves. [eLife 5:e23156](#) (2016).**  
*\*Corresponding authors.*
31. **Atwal G, Kinney JB. Learning quantitative sequence-function relationships from massively parallel experiments. [J Stat Phys 162\(5\):1203-1243](#) (2016).**
32. Sheu Y-J, Kinney JB, Stillman B. Concerted activities of Mcm4, Sld3 and Dbf4 in control of origin activation and DNA replication fork progression. [Genome Res 26:315-330](#) (2016).
33. **Kinney JB. Unification of field theory and maximum entropy methods for learning probability densities. [Phys Rev E 92:032107](#). (2015).**

34. Shi J, Wang E, Milazzo JP, Wang A, Kinney JB, Vakoc CR. Discovery of cancer drug targets by CRISPR-Cas9 screening of protein domains. [Nat Biotechnol 33\(6\):661-667](#) (2015).
35. Wang E, Kawaoka S, Roe J-S, Shi J, Hohmann AF, Xu Y, Bhagwat AS, Suzuki Y, Kinney JB, Vakoc CR. The transcriptional cofactor TRIM33 prevents apoptosis in B lymphoblastic leukemia by deactivating a single enhancer. [eLife 4:e06377](#) (2015).
36. Kinney JB. Estimation of probability densities using scale-free field theories. [Phys Rev E 90:011301\(R\)](#) (2014).
37. Kinney JB, Atwal GS. Parametric inference in the large data limit using maximally informative models. [Neural Comput 26\(4\):637-653](#) (2014).
38. Kinney JB, Atwal GS. Equitability, mutual information, and the maximal information coefficient. [Proc Natl Acad Sci USA 111\(9\):3354-3359](#) (2014).
39. Sheu Y-J, Kinney JB, Lengronne A, Pasero P, Stillman B. Domain within the helicase subunit Mcm4 integrates multiple kinase signals to control DNA replication initiation and fork progression. [Proc Natl Acad Sci USA 111\(18\):E1899-E1908](#) (2014).
40. Razo-Mejia M, Boedicker JQ, Jones D, Deluna A, Kinney JB, Phillips R. Comparison of the theoretical and real-world evolutionary potential of a genetic circuit. [Phys Biol 11\(2\):026005](#) (2014).
41. Weirauch MT, Cote A, Norel R, Annala M, Zhao Y, Riley TR, Saez-Rodriguez J, Cokelaer T, Vedenko A, Talukder S, DREAM5 Consortium, Bussemaker HJ, Morris QD, Bulyk ML, Stolovitzky G, Hughes TR. Evaluation of methods for modeling transcription factor sequence specificity. [Nat Biotechnol 31\(2\):126-134](#) (2013).
42. Melnikov A, Murugan A, Zhang X, Tesileanu T, Wang L, Rogov P, Feizi S, Gnirke A, Callan CG, Kinney JB, Kellis M, Lander ES, Mikkelsen TS. Rapid dissection and model-based optimization of inducible enhancers in human cells using a massively parallel reporter assay. [Nat Biotechnol 30\(3\):271-277](#) (2012).
43. Kinney JB, Murugan A, Callan CG, Cox EC. Using deep sequencing to characterize the biophysical mechanism of a transcriptional regulatory sequence. [Proc Natl Acad Sci USA 107\(20\):9158-9163](#) (2010).
44. Kinney JB. Biophysical models of transcriptional regulation from sequence data. [PhD dissertation, Princeton University](#) (2008).
45. Mustonen V, Kinney JB, Callan CG, Lässig M. Energy-dependent fitness: a quantitative model for the evolution of yeast transcription factor binding sites. [Proc Natl Acad Sci USA 105\(34\):12376-12381](#) (2008).
46. Kinney JB, Tkačik G, Callan CG. Precise physical models of protein-DNA interaction from high-throughput data. [Proc Natl Acad Sci USA 104\(2\):501-506](#) (2007).
47. Kinney JB, Maldacena J, Minwalla S, Raju S. An index for 4 dimensional super conformal theories. [Commun Math Phys 275\(1\):209-254](#) (2007).
48. Hines C, Childress L, Kinney JB, Sulzer M. Modeling of gravity-wave tail spectra in the middle atmosphere via numerical and Doppler-spread methods. [J Atmos Sol-Terr Phys 66:933-948](#) (2004).
49. Kinney JB, Mendell G.  $r$ -modes in accreting neutron stars with magnetoviscous boundary layers. [Phys Rev D 67:024032](#) (2003).

## Other Publications

1. Kinney JB, McCandlish DM. Massively parallel assays and quantitative sequence-function relationships. [Annu Rev Genom Hum G 20:99-127](#) (2019).
2. Sachan N, Miller M, Shirole NH, Senturk S, Corbo V, Kinney JB, Sordella R. Inducible CRISPR-based Genome Editing for the Characterization of Cancer Genes. [Genome Editing and Engineering: From TALENs, ZFNs and CRISPRs to Molecular Surgery](#), Cambridge University Press, p. 337-357 doi: <https://doi.org/10.1017/9781316756300.024> (2018).
3. Kinney JB. Mutual information: a universal measure of statistical dependence. [Biomed Computat Rev 10\(2\):33](#) (2014).